

CSA1613 - Data Warehousing and Data Mining

Q. Mythili

192411256

CO2 - ATQ :

1. a) Data Mining : It is The Process of finding useful Patterns and Knowledge from large amounts of data

Two Data Mining Tasks :

* Classification

* Clustering

KDD Steps :

→ Data Cleaning

→ Data Integration

→ Data Selection

→ Data Transformation

→ Data Mining

→ Pattern Evaluation

b) Data Cleaning :

Data cleaning is The Process of removing errors and improving data Quality.

Two Data Quality Issues :

* Missing Values

* Noisy data

Handling Missing Values :

* Replace with mean, median or mode

* Delete The record if necessary.

Handling Noisy data:

- * Binning
- * Smoothing

Q. a) Data Integration: It combines data from different sources into one dataset.

Challenges → Redundancy
→ Inconsistency

Handling Them:

- * Remove duplicate data
- * Standardize data formats

Ex:

DOB : 12/08/2005

2005-08-12

Converted into one common format.

b) Covariance Analysis: Covariance measures how 2 variables change together.

Two Covariance formulae:

Population Covariance
$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Interpretation:

- * Positive Covariance $\rightarrow$ Increase together
- * Negative Covariance $\rightarrow$ one increases while the other decreases.
- * Zero Covariance $\rightarrow$ independent.

3. a) Given :

A	B
2	5
3	8
5	10
4	11
6	14

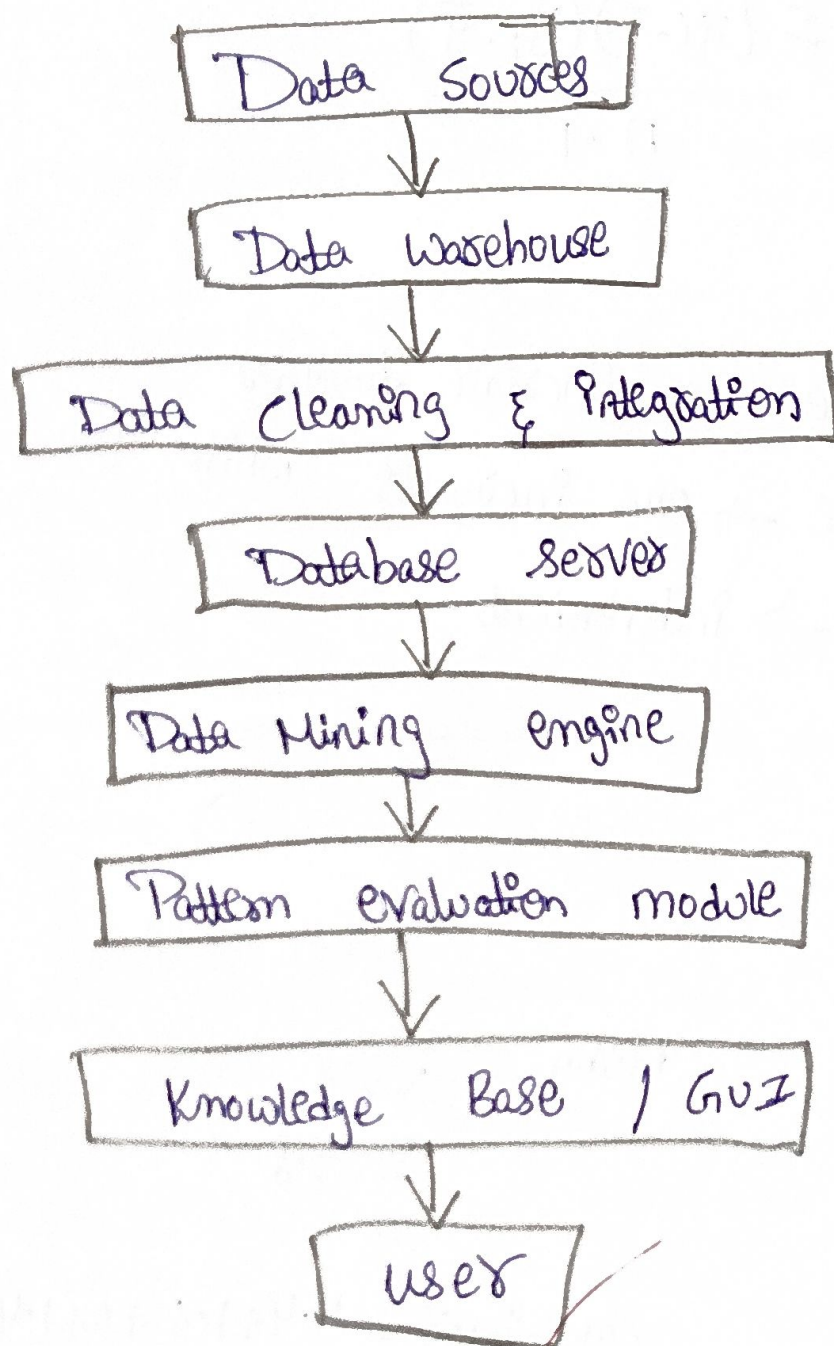
Mean $\bar{x} = 4$
 $\bar{y} = 9.6$

$$\begin{aligned}\text{Covariance} &= \frac{4 \cdot 4 + 1 \cdot 6 + 0 + 1 \cdot 4 + 8 \cdot 8}{5} \\ &= \frac{16.9}{5} \\ &= 3.24\end{aligned}$$

Therefore ;

- * Stocks A & B rise together
- * They are Positively related
- * They are not independent.

b) Architecture of Data Mining System:



4. a) Given;

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean} \rightarrow \bar{X} = 6.2$$

$$\bar{Y} = 5.8$$

Products:

x_i	y_i	$(x_i - \bar{x})(y_i - \bar{y})$
4	8	
5	3	-4.84
6	7	5.04
7	5	-0.24
9	6	-0.96
		0.44

$$\text{Sum} = -0.56$$

$$\text{Covariance} = \frac{-0.56}{5} = -0.112$$

b) Major Data Preprocessing steps:

- * Data Cleaning
- * Data Integration
- * Data Reduction
- * Data discretization.

Justification:

- Cleaning 1st removes errors
- Outlier removal Prevents bias
- Standardization ensures consistency
- Normalization improves mining
- Reduction decreases storage & computation.

Removing missing values → remove outliers → standardize

store dataset

← Reduce data ← Normalize data ← format

5.

a)

Equal-Frequency Binning:

- Divide data into equal-sized bins
- Replace values with the bin average
- This reduces noise in the data.

Given ages:

13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 1:

Mean = 18

Smoothed Bin:

18, 18, 18, 18, 18, 18, 18, 18, 18

22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 2:Mean ≈ 28

Smoothed Bin:

28, 28, 28, 28, 28, 28, 28, 28, 28

b) Data Discretization:

It converts continuous data into discrete intervals

Importance:

- Reduces complexity
- Removes noise
- Speeds up mining

Impact on Data Analysis:

- * Equal-width is easy but may create unbalanced bins
- * Equal-frequency provides balanced bins & often improves analysis for skewed datasets.